

Chapter 5. Machine Learning

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Today's topics

- Online Learning
- Online to Batch Conversion
- Support-Vector Machines
- VC-Dimension

Review

Binary classification problem : Using training examples S , classify examples in $\mathcal{X}$ into positive and negative examples.

$\mathcal{X}$: instance space (e.g. emails)

$S(\subseteq \mathcal{X})$: training examples (e.g. emails with correct classification)

$c^*(\subseteq \mathcal{X})$: target concept (e.g. spam emails)

$h(\subseteq \mathcal{X})$: hypothesis

$$\text{cf. } h(x) := \begin{cases} 1 & x \in h \\ -1(\text{or } 0) & x \notin h \end{cases}$$

Main results

Theorem 5.3 Let $\mathcal{H}$ be a hypothesis class and let ϵ and δ be greater than zero. If a training set S of size

$$n \geq \frac{1}{\epsilon}(\ln |\mathcal{H}| + \ln(1/\delta)),$$

is drawn from distribution D , then with probability greater than or equal to $1 - \delta$ every $h \in \mathcal{H}$ with true error $\text{err}_D(h) \geq \epsilon$ has training error $\text{err}_S(h) > 0$. Equivalently, with probability greater than or equal to $1 - \delta$, every $h \in \mathcal{H}$ with training error zero has true error less than ϵ .

cf.

$\text{err}_D(h) = \mathbb{P}_D[h \Delta c^*]$ (true error)

$\text{err}_S(h) = |S \cap (h \Delta c^*)|/|S|$ (training error)

Theorem 5.5 (Uniform convergence) Let $\mathcal{H}$ be a hypothesis class and let ϵ and δ be greater than zero. If a training set S of size

$$n \geq \frac{1}{2\epsilon^2}(\ln |\mathcal{H}| + \ln(2/\delta)),$$

is drawn from distribution D , then with probability greater than or equal to $1 - \delta$, every $h \in \mathcal{H}$ satisfies $|\text{err}_S(h) - \text{err}_D(h)| \leq \epsilon$.

Batch learning vs Online learning

In batch learning scenario...

- We are given a batch of data S
- S is drawn from a fixed probability distribution D on $\mathcal{X}$
- Objective : Find a hypothesis h that has a small true error
 $\text{err}_D(h) = \mathbb{P}_D[h \Delta c^*] = \mathbb{P}_{x \sim D}[h(x) \neq c^*(x)]$

In online learning scenario...

- Data are available in a sequential order
- There is no underlying distribution D
- Objective : Find a learning algorithm $\mathcal{A}$ that minimizes the total number of mistakes

Formal description

x_t : t -th example, h_t : t -th hypothesis

1. The algorithm is presented with an arbitrary example $x_t \in \mathcal{X}$ and is asked to make a prediction $h_t(x_t)$ of its label.
2. The algorithm is told the true label of the example $c^*(x_t)$ and is charged for a mistake if $c^*(x_t) \neq h_t(x_t)$

Remark. Intuitively, it is clear that the online learning will be much harder than the batch learning.

Online learning examples

1. Learning disjunctions(OR) problem

$$\mathcal{X} = \{0, 1\}^d, c^*(y_1, \dots, y_d) = y_{i_1} \vee \dots \vee y_{i_k}$$

Strategy

- Start with initial hypothesis $h_1(y) = y_1 \vee \dots \vee y_d$
- If $h_t(x_t) \neq c^*(x_t)$, remove from h_t any variable set to 1 in x_t and denote it by h_{t+1}

The above algorithm makes at most d mistakes in total on any series of examples consistent with a disjunction.

Theorem 5.9 For any deterministic algorithm $\mathcal{A}$ there exists a sequence of examples σ and disjunction c^* such that $\mathcal{A}$ makes at least d mistakes on sequence σ labeled by c^* .

Proof. $\sigma = \{e_1, \dots, e_d\}$, where $e_j = (0, \dots, 1^{j\text{-th}}, \dots, 0)$.

Define c^* such that $\mathcal{A}$ makes a mistake on every example, i.e. $h_t(e_t) \neq c^*(e_t)$.

This defines a legal disjunction. □

2. The halving algorithm

$\mathcal{H}$: hypothesis class(set of all possible hypothesis)

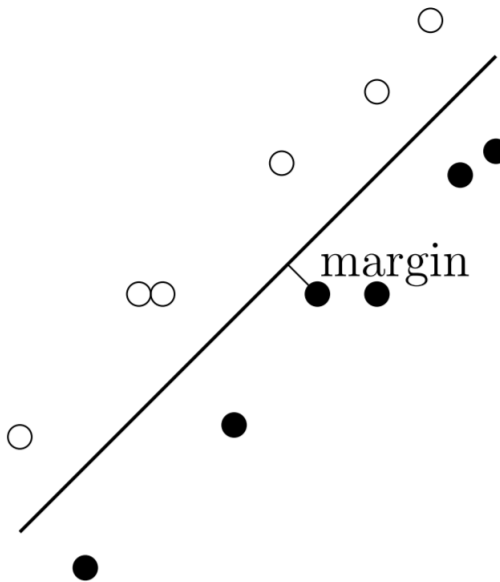
Strategy

- Start with $\mathcal{V}_1 = \mathcal{H}$
- Predict based on majority vote over $\mathcal{V}_t$, i.e.
$$h_t(x_t) = \begin{cases} 1 & |\{h \in \mathcal{V}_t : h(x_t) = 1\}| \geq |\mathcal{V}_t|/2 \\ -1 & |\{h \in \mathcal{V}_t : h(x_t) = 1\}| < |\mathcal{V}_t|/2 \end{cases}$$
- If $h_t(x_t) \neq c^*(x_t)$, drop the hypotheses with wrong predictions, i.e. $\mathcal{V}_{t+1} = \{h \in \mathcal{V}_t : h(x_t) = c^*(x_t)\}$

This algorithm guarantees to make at most $\log_2(|\mathcal{H}|)$ mistakes for a target $c^* \in \mathcal{H}$.

3. The perceptron algorithm and its extension

c^* can be described by a vector $\mathbf{w}^*$ such that $\mathbf{x}^T \mathbf{w}^* \geq 1$ for each positive example $\mathbf{x}$ and $\mathbf{x}^T \mathbf{w}^* \leq -1$ for each negative example $\mathbf{x}$. Equivalently, all examples are at distance at least $\gamma = 1/|\mathbf{w}^*|$ (margin of separation) from the separator.



Perceptron algorithm described in Section 5.2 also works well in the online setting.

The Perceptron Algorithm : Start with the all-zeroes weight vector $\mathbf{w} = 0$. Then, for $t = 1, 2, \dots$ do :

1. Given example $\mathbf{x}_t$, predict $\text{sgn}(\mathbf{x}_t^T \mathbf{w})$.
2. If the prediction was a mistake, then update :
 - (a) If $\mathbf{x}_t$ was a positive example, let $\mathbf{w} \leftarrow \mathbf{w} + \mathbf{x}_t$.
 - (b) If $\mathbf{x}_t$ was a negative example, let $\mathbf{w} \leftarrow \mathbf{w} - \mathbf{x}_t$.

Theorem 5.10 On any sequence of examples $\mathbf{x}_1, \mathbf{x}_2, \dots$, the perceptron algorithm makes at most $R^2 |\mathbf{w}^*|^2$ mistakes, where $R = \max_t |\mathbf{x}_t|$.

Idea of proof : For each mistake, $\mathbf{w}^T \mathbf{w}^*$ increases by at least 1 and $|\mathbf{w}|^2$ increases by at most R^2 . Use $\mathbf{w}^T \mathbf{w}^* \leq |\mathbf{w}| |\mathbf{w}^*|$.

Similar statement can be made even if the hyperplane $\mathbf{x}^T \mathbf{w}^*$ does not perfectly separate the positive and negative examples.

Define *hinge-loss* of $\mathbf{w}^*$ on $\mathbf{x}_t$ as

$$L_{\text{hinge}}(\mathbf{w}^*, \mathbf{x}_t) = \begin{cases} \max(0, 1 - \mathbf{x}_t^T \mathbf{w}^*) & \mathbf{x}_t \text{ is a positive example} \\ \max(0, 1 + \mathbf{x}_t^T \mathbf{w}^*) & \mathbf{x}_t \text{ is a negative example} \end{cases}.$$

Also, $L_{\text{hinge}}(\mathbf{w}^*, S) := \sum_{\mathbf{x}_t \in S} L_{\text{hinge}}(\mathbf{w}^*, \mathbf{x}_t)$.

Theorem 5.11 On any sequence of examples $S = \mathbf{x}_1, \mathbf{x}_2, \dots$, the perceptron algorithm makes at most

$$\min_{\mathbf{w}^*} (R^2 |\mathbf{w}^*|^2 + 2L_{\text{hinge}}(\mathbf{w}^*, S))$$

mistakes, where $R = \max_t |\mathbf{x}_t|$.

Online to batch conversion

An online learning algorithm can be converted to a batch learning algorithm as follows.

Theorem 5.12 (Online to Batch via Random Stopping) If an online algorithm $\mathcal{A}$ with mistake-bound M is run on a sample S of size M/ϵ and stopped at a random time between 1 and $|S|$, the expected error of the hypothesis h produced satisfies $\mathbb{E}[\text{err}_D(h)] \leq \epsilon$

Proof. Let X_t be the indicator random variable for the event that $\mathcal{A}$ makes a mistake on example $\mathbf{x}_t$. Since $\sum_{t=1}^{|S|} X_t \leq M$ for any set S , we certainly have that $\mathbb{E}[\sum_{t=1}^{|S|} X_t] \leq M$. Dividing both sides by $|S|$, we have

$$\frac{1}{|S|} \sum_{t=1}^{|S|} \mathbb{E}[X_t] \leq M/|S| = \epsilon.$$

Since the t -th example was randomly drawn from D , we have $\mathbb{E}[\text{err}_D(h_t)] = \mathbb{E}[X_t]$. □

However, Theorem 5.12 does not guarantee that every trial has a small true error. We need an alternative way of conversion.

Algorithm(Controlled Testing)

1. Test n_t random examples with hypothesis h_t .
- 2(a). If h_t predicts all example correctly, output h_t as our hypothesis and halt.
- 2(b). If h_t makes a mistake, give it to algorithm $\mathcal{A}$ to obtain new hypothesis h_{t+1} .

Choice of n_i 's

- $0 < \delta < 1$
- $\delta_i = \delta / (i + 2)^2$
- $n_i = \frac{1}{\epsilon} \log(\frac{1}{\delta_i})$

Theorem 5.13 (Online to Batch via Controlled Testing)

Let $\mathcal{A}$ be an online learning algorithm with mistake-bound M . Then this procedure will halt after $O(\frac{M}{\epsilon} \log(\frac{M}{\delta}))$ examples and with probability at least $1 - \delta$ will produce a hypothesis of error at most ϵ .

Support-vector machines

Recall the upper bound in Theorem 5.11 :

$$\min_{\mathbf{w}^*} (R^2 |\mathbf{w}^*|^2 + 2L_{\text{hinge}}(\mathbf{w}^*, S)).$$

Also, observe that

- minimizing $R^2 |\mathbf{w}^*|^2 \iff$ maximizing the margin of separation
- minimizing $L_{\text{hinge}}(\mathbf{w}^*, S) \iff$ minimizing the amount of “exceptional data”

Therefore, finding a “good” separating hyperplane $\mathbf{x}^T \mathbf{w}$ is equivalent to solving

$$\text{minimize } c|\mathbf{w}|^2 + \sum_i s_i$$

subject to $\mathbf{w} \cdot \mathbf{x}_i \geq 1 - s_i$ for all positive examples $\mathbf{x}_i$

$\mathbf{w} \cdot \mathbf{x}_i \leq -1 + s_i$ for all negative examples $\mathbf{x}_i$

$s_i \geq 0$ for all i .

Remark. c is determined empirically.

Lagrangian dual and kernel trick

Dual problem

$$\text{maximize } f(c_1, \dots, c_n) = \sum_{i=1}^n c_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i c_i y_j c_j (\mathbf{x}_i \cdot \mathbf{x}_j)$$

$$\text{subject to } \sum_{i=1}^n c_i y_i = 0$$

$$0 \leq c_i \leq \frac{1}{2n\lambda}$$

Kernel trick

$$\text{maximize } f(c_1, \dots, c_n) = \sum_{i=1}^n c_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i c_i y_j c_j k(\mathbf{x}_i, \mathbf{x}_j)$$

$$\text{subject to } \sum_{i=1}^n c_i y_i = 0$$

$$0 \leq c_i \leq \frac{1}{2n\lambda}$$

Vapnik-Chervonenkis (VC) dimension

Question : How can we measure the complexity of $\mathcal{H}$?

Answer : $|\mathcal{H}|$ (...?)

Recall the Theorem 5.3, and Theorem 5.5

Theorem 5.3 Let $\mathcal{H}$ be a hypothesis class and let ϵ and δ be greater than zero. If a training set S of size

$$n \geq \frac{1}{\epsilon} (\ln |\mathcal{H}| + \ln \frac{1}{\delta})$$

is drawn from distribution D , then with probability greater than or equal to $1 - \delta$ every h in $\mathcal{H}$ with true error $\text{err}_D(h) \geq \epsilon$ has training error $\text{err}_S(h) < 0$. Equivalently, with probability greater than or equal to $1 - \delta$, every $h \in \mathcal{H}$ with training error zero has true error less than ϵ .

Theorem 5.5 (Uniform convergence) Let $\mathcal{H}$ be a hypothesis class and let ϵ and δ be greater than zero. If a training set S of size

$$n \geq \frac{1}{2\epsilon^2}(\ln |\mathcal{H}| + \ln(2/\delta)),$$

is drawn from distribution D , then with probability greater than or equal to $1 - \delta$, every $h \in \mathcal{H}$ satisfies $|\text{err}_S(h) - \text{err}_D(h)| \leq \epsilon$.

What can we say if $|\mathcal{H}| = \infty$?

Observation : Two hypothesis $h_1, h_2 \in \mathcal{H}$ make same prediction on a training set S if $h_1 \cap S = h_2 \cap S$.

e.g. $\mathcal{H} = \{\text{axis-parallel rectangles}\}$, $S = \{\text{points on } \mathbb{R}^2\}$

Idea : Use “effective” hypothesis class to define its complexity!

Definition 5.3 Given a set S of examples and a concept class $\mathcal{H}$, let $\mathcal{H}[S] = \{h \cap S : h \in \mathcal{H}\}$. That is, $\mathcal{H}[S]$ is the concept class $\mathcal{H}$ restricted to the set of points S . For integer n and class $\mathcal{H}$, let $\mathcal{H}[n] = \max_{|S|=n} |\mathcal{H}[S]|$; this is called the **growth function** of $\mathcal{H}$.

e.g.

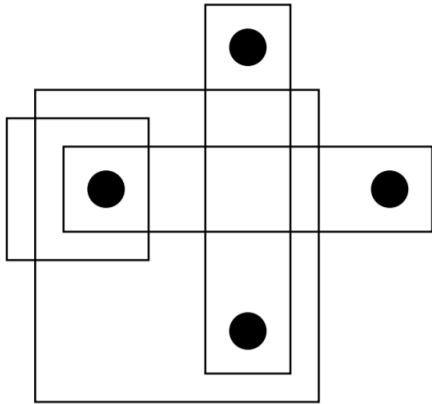
$$\mathcal{H} = \{\text{Axis-parallel rectangles}\} \Rightarrow \mathcal{H}[n] = O(n^4)$$

$$\mathcal{H} = \{\text{Intervals of the reals}\} \Rightarrow \mathcal{H}[n] = O(n^2)$$

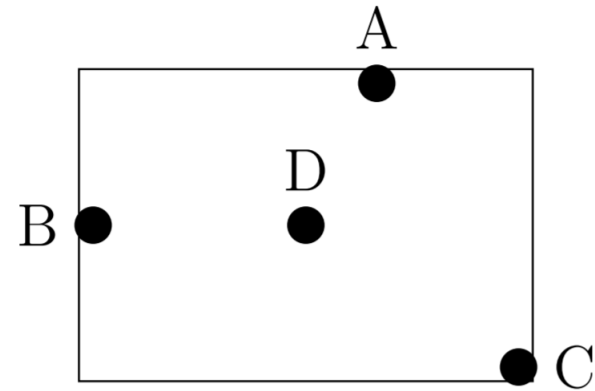
Sometimes, however, it might be hard to directly calculate the growth function of $\mathcal{H}$. VC-dimension provides a way to estimate the growth function.

Definition 5.1 Given a set S of examples and a concept class $\mathcal{H}$, we say that S is **shattered** by $\mathcal{H}$ if for every $A \subseteq S$ there exists some $h \in \mathcal{H}$ that labels all examples in A as positive and all examples in $S \setminus A$ as negative.

- S is shattered by $\mathcal{H}$ if $|\mathcal{H}[S]| = 2^{|S|}$.



(a)



(b)

- Points in (a) are shattered by axis-parallel rectangles.
- Points in (b) are not shattered by axis-parallel rectangles.

Definition 5.2 The **VC-dimension** of $\mathcal{H}$ is the size of the largest set shattered by $\mathcal{H}$.

- VC-dimension of $\mathcal{H}$ is the largest n such that $\mathcal{H}[n] = 2^n$.

e.g. VC-dimension of axis-parallel rectangles is four.

Theorem 5.16 (Sauer's lemma) If $\text{VCdim}(\mathcal{H}) = d$, then $\mathcal{H}[n] \leq \sum_{i=0}^d \binom{n}{i} := \binom{n}{\leq d} \leq (\frac{en}{d})^d$.

Proof. Induction on n . Since

$$\binom{n}{\leq d} = \binom{n-1}{\leq d} + \binom{n-1}{\leq d-1},$$

it suffices to prove that $|\mathcal{H}[S]| - |\mathcal{H}[S \setminus \{x\}]| \leq \binom{n-1}{\leq d-1}$ for a point $x \in S$.

Observe that $|\mathcal{H}[S]| - |\mathcal{H}[S \setminus \{x\}]| = |\mathcal{T}|$, where $\mathcal{T} = \{h \in \mathcal{H}[S] : x \in h \text{ and } h \setminus \{x\} \in \mathcal{H}[S]\}$.

If d' points are shattered by $\mathcal{T}$, $(d' \text{ points}) \cup \{x\}$ can be shattered by $\mathcal{H}$.

$\therefore \text{VCdim}(\mathcal{T}) \leq d - 1 \Rightarrow |\mathcal{T}| \leq \binom{n-1}{\leq d-1}$ by induction. □

$|\mathcal{H}|$ in Theorem 5.3 and Theorem 5.5 can be replaced by the growth function.

Theorem 5.3 Let $\mathcal{H}$ be a hypothesis class and let ϵ and δ be greater than zero. If a training set S of size

$$n \geq \frac{1}{\epsilon}(\ln |\mathcal{H}| + \ln(1/\delta)),$$

is drawn from distribution D , then with probability greater than or equal to $1 - \delta$ every $h \in \mathcal{H}$ with true error $\text{err}_D(h) \geq \epsilon$ has training error $\text{err}_S(h) > 0$. Equivalently, with probability greater than or equal to $1 - \delta$, every $h \in \mathcal{H}$ with training error zero has true error less than ϵ .

Theorem 5.14 (Growth function sample bound) For any class $\mathcal{H}$ and distribution D , if a training sample S is drawn from D of size

$$n \geq \frac{2}{\epsilon} [\log_2(2\mathcal{H}[2n]) + \log_2(1/\delta)]$$

then with probability $\geq 1 - \delta$, every $h \in \mathcal{H}$ with $\text{err}_D(h) \geq \epsilon$ has $\text{err}_S(h) > 0$ (equivalently, every $h \in \mathcal{H}$ with $\text{err}_S(h) = 0$ has $\text{err}_D(h) < \epsilon$).

Theorem 5.5 (Uniform convergence) Let $\mathcal{H}$ be a hypothesis class and let ϵ and δ be greater than zero. If a training set S of size

$$n \geq \frac{1}{2\epsilon^2}(\ln |\mathcal{H}| + \ln(2/\delta)),$$

is drawn from distribution D , then with probability greater than or equal to $1 - \delta$, every $h \in \mathcal{H}$ satisfies $|\text{err}_S(h) - \text{err}_D(h)| \leq \epsilon$.

Theorem 5.15 (Growth function uniform convergence) For any class $\mathcal{H}$ and distribution D , if a training sample S is drawn from D of size

$$n \geq \frac{8}{\epsilon^2}[\ln(2\mathcal{H}[2n]) + \ln(1/\delta)]$$

then with probability $\geq 1 - \delta$, every $h \in \mathcal{H}$ will have $|\text{err}_S(h) - \text{err}_D(h)| \leq \epsilon$.

Combining these theorems together, we have

Corollary 5.17 (VC-dimension sample bound) For any class $\mathcal{H}$ and distribution D , a training sample S of size

$$O\left(\frac{1}{\epsilon}[\text{VCdim}(\mathcal{H}) \log(1/\epsilon) + \log(1 - \delta)]\right)$$

is sufficient to ensure that with probability $\geq 1 - \delta$, every $h \in \mathcal{H}$ with $\text{err}_D(h) \geq \epsilon$ has $\text{err}_S(h) > 0$ (equivalently, every $h \in \mathcal{H}$ with $\text{err}_S(h) = 0$ has $\text{err}_D(h) < \epsilon$).

Thank you for your attention!